Towards Radar-Aware VLM-Based Autonomous Driving via mmWave Radar-Vision-Language Reasoning

Anonymous submission

Abstract

While vision-language models (VLMs) have triggered a shift toward language-driven autonomous driving, their reliance on purely visual inputs leaves them vulnerable to visually challenging conditions. Millimeter-wave (mmWave) radar provides a complementary modality with weather-invariant sensing as well as Doppler measurements, yet its noisy and sparse data structure makes it difficult to directly incorporate it into existing VLMs. To bridge this gap between physical sensor priors and semantic reasoning spaces, we propose **mmPointRet**, a question-guided retrieval framework. Rather than forcing the VLM to process raw point clouds, mmPointRet aligns natural language queries with their precise visual referents to isolate compact, task-relevant evidence. We further introduce **mmDrive-QA**, the first radar-vision-language question answering (QA) benchmark for decision-critical driving, comprising eight safety-critical edge cases and four reasoning categories. Our evaluation on mmDrive-QA shows that mmWave radar provides complementary physical cues for autonomous driving, while mmPointRet further improves performance by filtering redundant radar points and preserving target-relevant evidence. Further analysis reveals that Doppler and intensity cues from mmWave radar provide complementary benefits for motion-related and visually challenging questions.

Introduction

Vision-language models (VLMs) (Alayrac et al. 2022; Li et al. 2024) have demonstrated significant potential in autonomous driving, advancing beyond conventional perception pipelines to enable scene understanding, explainable decision-making, and language-guided planning (Liu et al. 2025). Despite these capabilities, VLM-driven autonomous agents predominantly rely on visual inputs, suffering from vulnerability when camera observations are degraded, such as adverse weather, poor lighting, or severe occlusion (Gu et al. 2025; Choi et al. 2025). To enhance robustness, LiDAR sensors are often employed due to their accurate 3D geometric perception, but their high cost and computational demands limit widespread adoption (Cui et al. 2024). mmWave radar, in contrast, provides robust sensing under poor visibility while directly capturing physical information such as range and Doppler velocity (Wolters et al. 2025). These characteristics motivate the integration of radar with VLMs for robust and physically grounded autonomous driving.

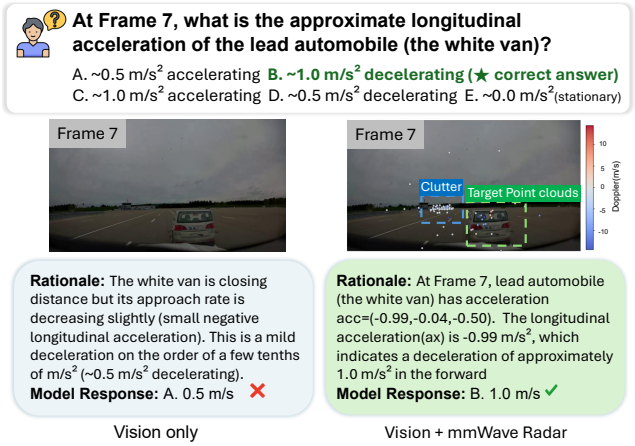


Figure 1: Effect of mmWave radar cues on language-based driving QA. Vision-only GPT-5mini fails to infer the acceleration, while adding radar point clouds and Doppler cues enables correct motion-aware reasoning.

As illustrated in Fig. 1, incorporating radar points into a VLM (GPT-5 mini) can improve question-answering (QA) performance in scenarios where vision alone struggles, such as inferring the longitudinal acceleration of a lead vehicle. However, realizing this multi-modal benefit introduces a fundamental representational challenge: unlike dense visual pixels, raw radar point clouds are unstructured, sparse, and frequently corrupted by noise and clutter reflections. Namely, directly feeding the raw/unfiltered radar point cloud not only induces substantial computational overhead by generating excessively long token sequences but also distracts the model with question-irrelevant clutter reflections (right image in Fig. 1). In addition, this context bottleneck is especially severe for compact VLMs with limited processing windows, making raw radar inputs difficult to process properly.

To enable efficient and physically grounded radar-vision-language reasoning, we propose a novel framework centered on *mmPointRet*, a question-guided mmWave radar point retriever. As illustrated in Fig. 3, mmPointRet dynamically

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identifies target-relevant spatial regions by aligning natural language queries with their corresponding visual referents. By extracting the subset of radar returns associated with the queried object, this approach eliminates clutter and distills the physical signal into a compact subset. The selected radar points are then converted into a language-compatible representation via radar-specific post-processing and provided to a downstream VLM together with visual observations and the question. To the best of our knowledge, this is the first attempt to ground VLM-based driving reasoning using question-guided radar retrieval. In particular, mmPointRet operates in a model-agnostic manner, allowing it to be integrated with various off-the-shelf VLMs alongside other modalities without requiring architectural modifications.

Meanwhile, another key bottleneck for advancing radar-aware VLMs is the complete absence of QA benchmarks to verify whether radar modality can actually benefit language models in autonomous driving. To systematically evaluate this radar-vision-language reasoning, we introduce *mmDrive-QA*, the first radar-vision-language QA benchmark for decision-critical autonomous driving scenarios (Tab. 1). The benchmark is constructed using synchronized vision, radar, and LiDAR data, where LiDAR serves as an auxiliary reference for annotation and evaluation. It covers eight challenging driving scenarios and includes language-based question answering tasks across four reasoning categories: perception, anticipation, reasoning, and planning. mmDrive-QA enables a comprehensive evaluation of whether language-driven driving agents can effectively leverage radar and vision cues for robust and explainable decision-making.

Our contributions are summarized as follows:

- We introduce a radar-vision-language reasoning framework for autonomous driving QA that enables VLMs to leverage mmWave radar cues along with visual observations for robust and explainable decision-making.
- We propose mmPointRet, a question-guided mmWave radar point retriever that selectively extracts target-relevant radar points from clutter and redundant radar point clouds, enabling scalable and efficient VLM-based reasoning in a model-agnostic manner.
- We introduce mmDrive-QA, the first radar-vision-language QA benchmark for decision-critical autonomous driving, covering eight challenging scenarios and four language-based driving-judgment tasks.

Related Work

Radar-Vision Fusion for Autonomous Driving. Radar and vision have been studied as complementary sensing modalities for autonomous driving (Nabati and Qi 2021; Kim et al. 2023). Early methods associated radar measurements with image-based object centers through frustum-based proposals (Qi et al. 2018; Nabati and Qi 2021) to improve 3D object detection and velocity estimation. Recent approaches have further explored radar-camera fusion in bird's-eye-view (BEV) representations and multi-view feature spaces (Kim et al. 2023; Lin et al. 2024). More recent works extend this direction to query-based fusion, 4D radar-camera cross-view fusion, and contrastive radar representation learning (Chu et al. 2025; Zhong et al. 2025; Hao et al. 2024). Despite these advances, existing radar-vision studies focus on detection-oriented perception, including 3D object detection, tracking, BEV perception, and calibration. They do not investigate how radar-derived geometric and motion-aware cues can support high-level language reasoning, risk assessment, or decision-making. Our work differs from prior radar-camera fusion studies by integrating radar and vision into a language-driven autonomous driving framework.

Benchmark	Input			Tmp	Edge	Kin.	Task
	T	V	R				
BDD-X (2018)	✓	✓	×	✓	×	×	cap.
DRAMA (2023)	✓	✓	×	✓	✓	×	cap.
NuScenes-QA (2024)	✓	✓	×	×	×	×	QA
DriveLM (2024)	✓	✓	×	×	×	×	QA
DriveBench (2025)	✓	✓	×	×	×	×	QA
Talk2Radar (2025b)	✓	×	✓	×	×	×	det.
WaterVG (2025a)	✓	✓	✓	×	×	×	det.
mmDrive-QA (Ours)	✓	✓	✓	✓	✓	✓	QA

Table 1: Positioning of mmDrive-QA. **T/V/R**: text, video, and radar input; **Tmp**: temporal sequence input; **Edge**: curated safety-critical edge cases; **Kin.**: kinematic quantities are the answer, not a premise given in the prompt. **Task**: captioning (cap.), detection (det.), or QA.

Language-Driven Autonomous Driving. Recent studies have explored VLMs for language-driven autonomous driving, aiming to move beyond conventional perception pipelines by enabling scene understanding, QA, explainable reasoning, and language-guided planning. For example, VLP (Pan et al. 2024) incorporates language models into driving planning as an intermediate reasoning interface, and recent works explore human-centered LLM-based decision-making with feedback and RAG-based policy retrieval (Ma et al. 2024). Beyond methodological advances, several benchmarks formulate driving as a language task (Tab. 1): BDD-X (Kim et al. 2018) and DRAMA (Malla et al. 2023) supervise free-form text explaining an ego action or a perceived risk, while DriveLM (Sima et al. 2024), NuScenes-QA (Qian et al. 2024), and DriveBench (Xie et al. 2025) pose QA over camera keyframes drawn from the normal driving distribution of their source dataset. However, these benchmarks remain vision-centric, leaving radar-specific cues such as range and Doppler velocity outside the reasoning process.

Radar-Language Models. Bridging language with radar, in contrast, remains confined to retrieval and detection. RSLM (Pushkareva et al. 2024) retrieves scenes from radar spectra by free text, Talk2Radar (Guan et al. 2025b) grounds referring expressions in 4D radar, and WaterVG (Guan et al. 2025a) extends this to radar and vision jointly. Crucially, the two radar-language benchmarks state the kinematics in the prompt as a *premise* for locating a referent—a target's velocity and depth in Talk2Radar, its distance and motion trend in WaterVG—so the model matches a description rather than deriving a quantity, and a box cannot express a deceleration rate or a time-to-collision in the first place. Our mmDrive-

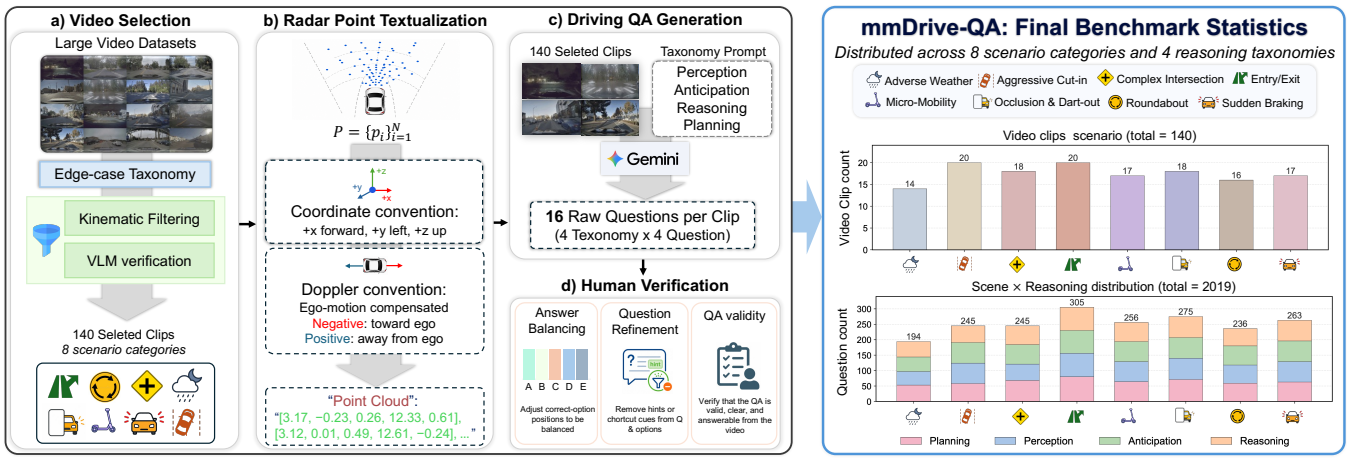


Figure 2: **mmDrive-QA curation pipeline.** The benchmark is constructed through four main stages: video selection, radar point textualization, driving QA generation, and human verification. The final benchmark contains 2,019 multiple-choice QAs distributed across eight scenario categories and four reasoning categories.

QA instead curates safety-critical clips, supplies radar as raw returns, and makes the kinematic quantity the answer.

mmDrive-QA Benchmark

Despite the potential of mmWave radar for robust autonomous driving, a fundamental bottleneck in developing radar-aware language models is the absence of paired radar-vision-language question-answering datasets in complex driving scenarios. To overcome this gap, we introduce *mmDrive-QA*, the first radar-vision-language QA benchmark tailored for decision-critical driving. As shown in Fig. 2, the benchmark construction consists of four main stages: video selection, radar point textualization, driving QA generation, and human verification, followed by dataset statistics.

Video Selection. We curate our video subset from the NVIDIA PhysicalAI-Autonomous-Vehicles dataset (NVIDIA Corporation 2025), which provides multimodal data including driving videos, LiDAR, radar, and ego-motion signals. Since the source dataset predominantly consists of normal driving scenarios, we employ a two-stage filtering pipeline to identify critical events. First, we apply kinematic filtering using ego-motion signals and object tracks to capture anomalous motion patterns and interactions. However, since kinematic cues alone cannot fully capture the contextual criticality of a scene, we further leverage a VLM to verify whether each candidate accurately aligns with the intended scenario category and to assess its severity. This process yields a balanced set of highly critical, verified clips covering categories such as *micro-mobility*, *complex intersections*, *roundabouts*, *occlusion and dart-out*, *aggressive cut-in*, *sudden braking*, *entry/exit*, and *adverse weather*.

Radar Point Textualization. To enable off-the-shelf VLMs to consume radar point clouds without architectural modifications, we convert them into a structured textual format. For each video frame, the corresponding radar point cloud $\mathbf{P}$ is defined as a set of points, $\mathbf{P} = \{\mathbf{p}_i\}_{i=1}^{N}$, where N is the number of points and each point $\mathbf{p}_i = (x_i, y_i, z_i, d_i, s_i)$ contains 3D coordinates in meters, Doppler velocity in m/s, and intensity (radar cross section in dBsm). All points are expressed in the ego-vehicle coordinate frame ($+x$: forward, $+y$: left, $+z$: upward), and we use ego-motion-compensated Doppler velocity, where negative values indicate motion toward the ego vehicle. Each point is serialized into a text sequence, allowing the VLM to reason about radar-derived spatial and physical cues alongside visual context. Since raw radar returns introduce a high density of numeric tokens per scene, naive serialization leads to excessive token lengths. We retain the union of the top-100 points by absolute Doppler and the top-100 points by intensity, discarding all other points.

Driving QA Generation. For each selected clip, we generate driving-oriented QA pairs using Gemini 2.5 Flash (Google for Developers 2025). Specifically, we produce 16 raw questions per clip, evenly distributed across four key reasoning categories, *perception*, *anticipation*, *reasoning*, and *planning*, by providing tailored prompt instructions for each taxonomy. These categories evaluate whether a model can identify the states and locations of surrounding objects (*perception*), forecast their future trajectories (*anticipation*), infer the underlying causes of critical events (*reasoning*), and determine the optimal ego-vehicle action (*planning*). Each clip is sampled into 20 frames at 1 fps, and the 16 questions are anchored to different key frames within it. Each item is formatted as a multiple-choice question with five candidate options, including one ground-truth answer and a detailed rationale to support subsequent human verification.

Human Verification. Finally, we conduct a three-step human verification to filter out ambiguous QA pairs and to prevent shortcuts where questions could be solved without relying on visual or radar evidence. First, we perform *answer balancing* to ensure an even distribution across the five choices. Second, we execute *question refinement* to remove textual hints and prevent shortcuts, including distractor leak-

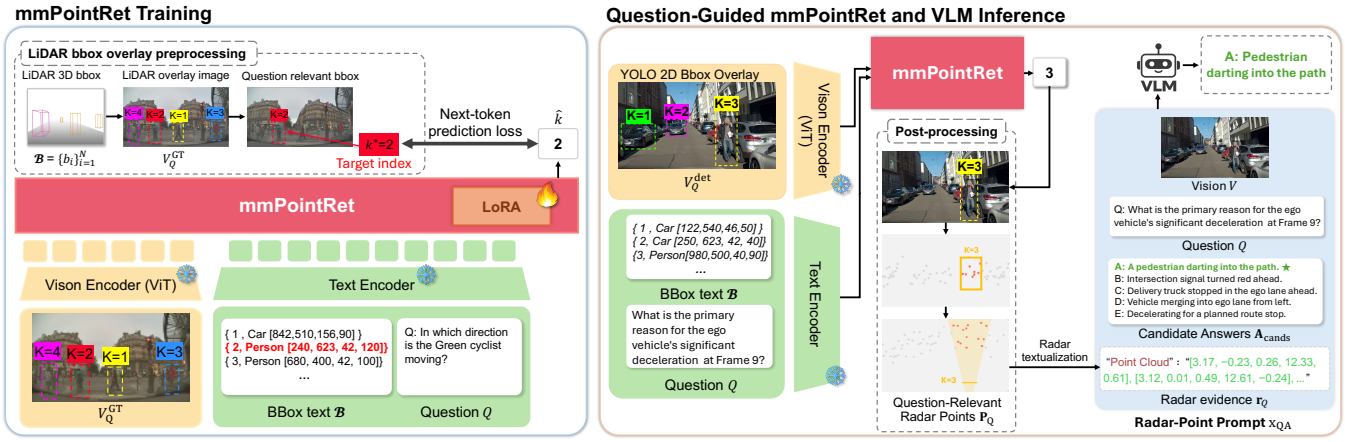


Figure 3: **Left:** LoRA-tuning of mmPointRet supervised by the LiDAR-derived target index k^* among N candidate boxes B. **Right:** At inference, YOLO-detected boxes replace LiDAR; the predicted box $b_{\hat{k}}$ filters the radar cloud into question-relevant points $\mathbf{P}_Q$, which form the radar prompt $\mathbf{x}_{QA}$ fed to the VLM with image V.

age, mutual-exclusivity shortcuts, and stem leakage. Third, we carry out a *QA validity* where annotators check whether each QA pair is valid, unambiguous, and requires scene-level understanding. Following the validity check, three non-author annotators independently reviewed all 2,019 items against the video and sensor evidence, flagging any item whose gold answer was inconsistent with that evidence or whose question or options leaked the correct choice The final verified subset forms *mmDrive-QA*, a benchmark designed to evaluate radar-vision-language reasoning in decision-critical driving scenarios.

Benchmark Statistics. After QA generation and human verification, the final mmDrive-QA bench consists of 2,019 human-verified multiple-choice QAs (Fig. 2). The questions cover eight safety-critical driving scenario categories and four reasoning categories, enabling a comprehensive evaluation of radar-vision-language reasoning. As summarized on the right of Fig. 2, the final set is broadly distributed across scenarios and reasoning types. See Appendix for details.

Method

Our framework is designed to facilitate radar-centric multimodal reasoning in autonomous driving by seamlessly integrating raw radar returns with video and text. Formally, given a serialized radar point cloud $\mathbf{P}$, a video sequence V, a question Q, and a set of candidate answers $\mathbf{A}_{\text{cands}}$, a VLM $\mathcal{M}$ infers the ground-truth answer A as follows:

$$A = \mathcal{M}(V, \mathbf{P}, Q, \mathbf{A}_{\text{cands}}). \tag{1}$$

A major obstacle is that integrating raw radar inputs into the language model is challenging because radar point clouds lack explicit object semantics and typically contain substantial noise and question-irrelevant clutter. Providing all radar points to $\mathcal{M}$ introduces question-irrelevant evidence and increases inference overhead. To mitigate this issue, question-guided radar point selection is required to retain question-relevant physical evidence while reducing irrelevant radar inputs before multimodal reasoning. Given Q and radar point cloud $\mathbf{P}$, the retriever Ret selects a question-relevant subset of radar points $\mathbf{P}_Q \subseteq \mathbf{P}$ as $\mathbf{P}_Q = \mathrm{Ret}(\mathbf{P}, Q)$. The overall radar-vision-language reasoning process is formulated as:

$$A = \mathcal{M}(V, \mathbf{P}_Q, Q, \mathbf{A}_{\text{cands}}). \tag{2}$$

Question-Guided Radar Point Retrieval

To efficiently extract $\mathbf{P}_Q$, we propose mmPointRet, a question-guided mmWave radar point retriever, as illustrated in Fig. 3. mmPointRet is designed to be integrated with VLMs in a model-agnostic manner. To this end, $\mathbf{P}_Q$ extracted by mmPointRet is converted into a structured textual format and provided together with visual observations and textual QA inputs. The detailed training objective and inference-time retrieval are formalized below.

Training Objectives. To provide reliable supervision for question-relevant grounding, we leverage LiDAR-derived 3D bounding boxes as high-fidelity geometric anchors during training. During the construction of *mmDrive-QA*, each question is annotated with its target object, allowing us to identify the corresponding LiDAR 3D box among multiple objects in the scene. For each training sample, we project LiDAR-derived 3D object boxes onto the anchor camera frame V_Q^{GT} to obtain candidate boxes $\mathcal{B} = \{b_i\}_{i=1}^N$ in pixel coordinates.

We instantiate mmPointRet with Qwen2.5-VL-3B-Instruct (Bai et al. 2025a), a lightweight VLM backbone, by attaching a LoRA adapter (Hu et al. 2022) to its language-model attention projections. Let p_θ denote the LoRA-adapted token-generation distribution of mmPointRet, where θ corresponds to the trainable LoRA parameters. Using the candidate boxes with V_Q^{GT} and the question Q, mmPointRet predicts the index $\hat{k}$ of the box corresponding to the question-relevant object:

$$\hat{k} = \underset{k \in \{1,\dots,N\}}{\arg\max}\ p_\theta(k \mid V_Q^{\mathrm{GT}}, Q, \mathcal{B}). \tag{3}$$

The ground-truth label k^* is the index of the target object's projected LiDAR box among the candidates. Because mm-

PointRet generates this index as text, we convert k^* into a target token sequence $y^* = \mathrm{Tok}(\mathrm{str}(k^*))$ and train the model with a next-token prediction objective:

$$\mathcal{L}_{\mathrm{ret}} = -\sum_{t \in \mathcal{Y}} \log p_\theta(y_t^* \mid V_Q^{\mathrm{GT}}, Q, \mathcal{B}, y_{<t}^*), \tag{4}$$

where $\mathcal{Y}$ denotes the output token positions corresponding to the tokenized target index y^*. Note that LiDAR-derived boxes are used only to obtain training supervision and are not required at inference time. mmPointRet is trained on referent-selection questions built from clips disjoint from those of *mmDrive-QA*. See Appendix for details.

Inference-time Retrieval. At inference time, mmPointRet retrieves question-relevant radar points without using LiDAR. Specifically, an off-the-shelf detector, such as YOLO11x (Jocher and Qiu 2024), first produces candidate boxes $\mathcal{B}$ from V_Q^{det}, and the trained mmPointRet selects the question-relevant box $b_{\hat{k}}$. Each radar point is projected onto the image plane, and only the points whose projections fall inside $b_{\hat{k}}$ are retained:

$$\mathbf{P}_Q = \{\mathbf{p}_i \in \mathbf{P} \mid \Pi(\mathbf{p}_i) \in b_{\hat{k}}\}, \tag{5}$$

where $\Pi(\cdot)$ denotes the radar-to-image projection function. This selection defines an angular wedge along the viewing directions covered by the selected 2D box, rather than a depth-bounded 3D region. Although the selected wedge may contain points at different depths, it provides compact radar cues associated with the visual referent. Despite the background clutter inherently introduced by this depth ambiguity, the drastically reduced point count avoids context bottlenecks and allows the VLM to implicitly filter out residual noise.

VLM Integration with mmPointRet

After retrieving question-relevant radar points with mmPointRet, we construct a radar-augmented prompt for VLM-based QA. The retrieved radar subset $\mathbf{P}_Q$ is converted into textual radar evidence $\mathbf{r}_Q$ using the structured radar textualization format. The final prompt $\mathbf{x}_{\mathrm{QA}}$ combines the visual observation V, question Q, candidate answers $\mathbf{A}_{\mathrm{cands}}$, and $\mathbf{r}_Q$. The instruction prompt also includes a brief description of the radar coordinate system and the Doppler sign convention, enabling the VLM to interpret the physical meaning of radar attributes. Given the resulting QA input $\mathbf{x}_{\mathrm{QA}}$, the VLM $\mathcal{M}$ generates a response with a brief chain-of-thought rationale, from which the final answer A is extracted. Because radar information is provided through a text-based interface, our method can be integrated with both closed-source and open-source VLMs without modifying their architectures.

Experiments

We examine whether mmWave radar cues improve VLM-based autonomous driving QA and analyze the efficiency of our proposed pipeline with mmPointRet.

Baselines. We evaluate both closed-source and open-source VLMs under three input settings: text with vision (`T+V`), text with vision and cap-filtered radar ($\texttt{T+V+R}_{\texttt{cap}}$), and text with vision and mmPointRet-retrieved radar (`ours`, $\texttt{T+V+R}_{\texttt{ret}}$). Here, $\texttt{R}_{\texttt{cap}}$ denotes the per-frame cap baseline, which takes the union of the top-100 points by absolute Doppler and the top-100 points by intensity, while $\texttt{R}_{\texttt{ret}}$ denotes the mmPointRet-retrieved radar described in the Method section. For closed-source models, we use GPT-5 and GPT-5-mini (OpenAI 2025), and Gemini 2.5 Flash (Google for Developers 2025). For open-source models, we evaluate Qwen3-VL (Bai et al. 2025b) at 32B and 8B scales, Qwen2.5-VL (Bai et al. 2025a) at 72B, 32B, and 7B scales, and LLaVA-Video (Zhang et al. 2024) at 7B. This setup allows us to analyze how the contribution of radar evidence and our proposed mmPointRet varies across different input conditions, VLM scales, and architectures.

Evaluation. We evaluate closed-set QA performance using accuracy over five options. A prediction is considered correct if the selected answer A matches the ground-truth answer A^* among the candidate set $\mathbf{A}_{\mathrm{cands}}$.

Main Results

Quantitative Results. Tab. 2 shows two consistent patterns. First, closed-source models (GPT-5, GPT-5-mini, Gemini 2.5 Flash) all benefit from adding radar: $\texttt{T+V+R}_{\texttt{cap}}$ consistently outperforms `T+V`. In contrast, for open-source models, adding the cap-filtered radar yields results comparable to or slightly worse than `T+V`. This contrast suggests that simply adding dense radar inputs is not always optimal, as the cap-filtered radar produces long token sequences that contain many noisy or question-irrelevant points. By selectively retaining target-relevant radar points, our mmPointRet-based pipeline consistently improves overall accuracy over $\texttt{T+V+R}_{\texttt{cap}}$ across both closed-source and open-source models, demonstrating that question-guided radar retrieval is a more effective way to incorporate mmWave radar into VLM-based driving QA. Among closed-source models, GPT-5 attains the highest accuracy under both $\texttt{T+V+R}_{\texttt{cap}}$ (56.61%) and our mmPointRet setting (56.91%), with performance saturating near the top. A larger improvement is observed for GPT-5-mini (from 51.86% to 52.85%) and Gemini 2.5 Flash (from 51.26% to 53.74%), suggesting that mmPointRet helps more when the base model does not already exploit dense radar prompts well. The effect becomes more pronounced for smaller open-source models. For Qwen3-VL-32B, mmPointRet improves performance from 43.54% to 47.75%, and for Qwen2.5-VL-32B from 40.61% to 43.14%. Most notably, Qwen2.5-VL-7B suffers a substantial drop with the $\texttt{T+V+R}_{\texttt{cap}}$ input, achieving only 10.65%, likely because the model cannot effectively process the long radar prompt and loses critical visual information. With mmPointRet, its performance recovers to 37.05%, demonstrating that compact, question-guided radar retrieval is especially beneficial for resource-constrained VLMs. Across the taxonomies, the Anticipation category exhibits the most substantial gains over the `T+V` baseline, improving by up to 18.55 pp (Gemini 2.5 Flash) and 12.30 pp (GPT-5). Since predicting dynamic behaviors from images alone is inherently difficult, the physical cues from mmWave radar effectively compensate for this limitation. Scene-level results are detailed in Appendix.

Input	Taxonomy	Closed-source			Open-source					
		GPT-5		Gemini 2.5	Qwen3		Qwen2.5			LLaVA
		base	mini	Flash	32B	8B	72B	32B	7B	7B
T+V	Perception	39.16	39.16	34.21	**38.35**	**35.54**	**37.35**	32.33	28.92	30.32
	Anticipation	39.65	38.48	30.08	37.50	**28.32**	35.74	33.20	27.15	**29.88**
	Reasoning	64.10	58.01	61.46	55.38	52.13	**58.62**	49.09	**44.22**	**48.07**
	Planning	57.17	54.65	57.75	56.20	52.52	52.33	48.64	43.80	44.57
	Overall	49.98	47.55	45.81	46.85	42.10	45.96	40.81	36.01	38.19
T+V+R_{cap}	Perception	**46.59**	41.57	40.56	32.53	28.92	35.14	30.92	10.64	30.32
	Anticipation	**52.54**	48.24	42.35	28.71	22.27	35.74	31.05	9.18	25.78
	Reasoning	**69.37**	**60.45**	62.47	56.39	53.35	57.40	50.71	10.55	43.61
	Planning	58.14	57.17	**59.88**	**56.59**	**53.29**	52.33	49.81	12.21	**47.67**
	Overall	56.61	51.86	51.26	43.54	39.43	45.12	40.61	10.65	36.85
T+V+R_{ret} (Ours)	Perception	45.98	**43.57**	**41.77**	36.75	32.53	36.75	**32.93**	**29.52**	**32.93**
	Anticipation	51.95	**49.22**	**48.63**	**39.65**	28.12	**45.31**	**34.57**	**30.66**	27.54
	Reasoning	68.97	59.63	**64.91**	**59.43**	**56.39**	55.98	**53.96**	44.02	46.45
	Planning	**60.85**	**58.91**	59.69	55.23	52.91	**53.10**	**51.16**	**43.99**	46.90
	Overall	**56.91**	**52.85**	**53.74**	**47.75**	**42.45**	**47.80**	**43.14**	**37.05**	**38.43**

Table 2: Quantitative evaluation results on the mmDrive-QA benchmark. Input modalities include T (Text), V (Vision), R_{cap} (per-frame cap), and R_{ret} (Ours). **Bold** and underline indicate the best and second-best performances.

	Method	Acc	#Pts	Rec.	Prec.	F1
(a)	LiDAR GT	42.72	4.4	Ref.	Ref.	Ref.
	R_{cap}	38.05	169.6	70.24	1.82	3.55
	mmPointRet (Ours)	**39.04**	**26.6**	**65.73**	**10.82**	**18.57**
(b)	YOLO11x wedge	37.62	36.0	45.80	5.57	9.93
	mmPointRet (Ours)	**39.04**	**26.6**	**65.73**	**10.82**	**18.57**

Table 3: (a) Radar point selection strategies and (b) retrievers on the 707-QA pool (Qwen2.5-VL-72B). Acc: QA accuracy. #Pts: cited-frame mean points per frame. Rec./Prec./F1: micro-averaged against the LiDAR GT point set, which the LiDAR-oracle retriever attains by construction.

Qualitative Results. Our qualitative cases (Fig. 4) illustrate why retriever-gated radar is necessary. Vision alone fails on metric questions because monocular cues compress depth and speed, leading the model to underestimate distances and relative velocities. Radar alone fails for a complementary reason. Without visual grounding, the model cannot identify which point cluster corresponds to the question's referent, so even accurate measurements cannot be tied to the right object. Vision + full-Radar provides both signals, but leaves the VLM to identify the referent among many noisy radar returns, which becomes the dominant source of fusion errors. mmPointRet removes this bottleneck. Before inference, each radar return is linked to the retriever's bounding box of the natural-language referent. The VLM thus receives metric evidence already associated with the queried object, producing consistent gains on questions whose answers depend on precise depth, distance, or velocity.

In-depth Analysis of the mmPointRet Framework

We further validate the efficacy of our mmPointRet from three different perspectives. First, we investigate the retrieval performance of mmPointRet itself: how well our approach narrows down the raw point clouds to select only the target-aware points compared to LiDAR GT (Tab. 3(a)). Second, we examine whether these retrieved points actually lead to accuracy gain via comparison with all-detected points (Tab. 3(b)). Finally, we investigate the depth ambiguity introduced by our 2D wedge, and compare against a retriever supervised without LiDAR (Tab. 4).

Method	Acc	#Pts	Rec.	Prec.	F1
Depth Anything V2	41.65	0.19	2.03	47.73	3.90
DINORet	46.30	—	45.83	8.97	15.00
mmPointRet (Ours)	**47.80**	**26.6**	**65.73**	**10.82**	**18.57**

Table 4: Comparison of monocular depth-based retrieval and LiDAR-free training. Acc is measured on the 2,019-QA benchmark, while #Pts and Rec./Prec./F1 are computed on the 707-QA retrieval subset.

Motion-related questions			
Subset	**All Attr.**	**w/o D**	**w/o D, I**
Numeric	40.41	35.23 (−5.18)	34.72 (−5.70)
Qualitative	51.22	47.80 (−3.42)	42.44 (−8.78)
Visibility-limited scenes			
Subset	**All Attr.**	**w/o I**	**w/o D, I**
Adv. Weather	60.31	55.67 (−4.64)	54.64 (−5.67)
Micro-Mob.	50.00	45.70 (−4.30)	48.83 (−1.17)

Table 5: Ablation of radar attributes on GPT-5-mini. D: Doppler, I: intensity. Parentheses: change from All Attr. Each group ablates the attribute expected to matter for it.

Retrieval Performance of mmPointRet. To compare selection strategies, we evaluate the LiDAR GT set (points inside the queried object's LiDAR 3D box), the R_{cap} baseline of our T+V+R_{cap} setting, and mmPointRet. The analysis is conducted on the 707-selected QA pool out of the whole

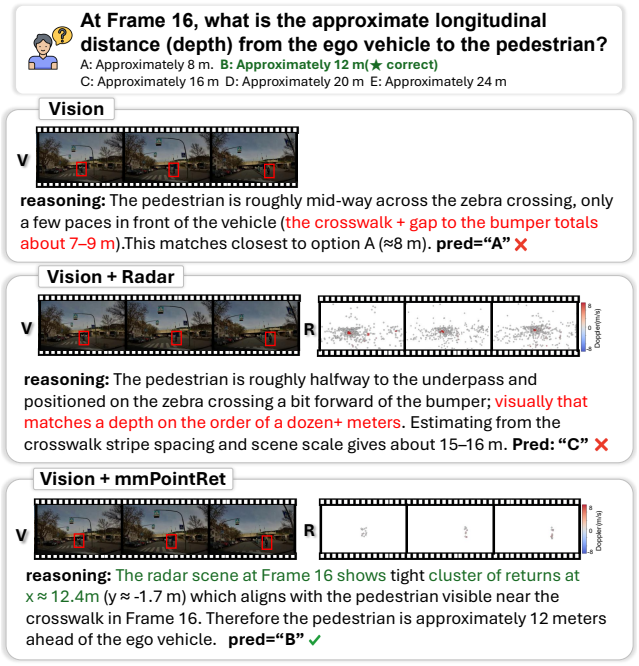


Figure 4: **Qualitative comparison on a depth-estimation question.** Radar panels are colored by ego-motion-compensated Doppler (blue: approaching, red: receding).

2,019 samples, whose questions target an object carrying a LiDAR 3D box at the cited frame. The raw radar cloud contains 508.5 points per frame on average, while only 4.4 fall inside the LiDAR target region. Compared with $\mathrm{R}_{\mathrm{cap}}$, mmPointRet uses 6.4× fewer points yet improves precision by 5.9× and F1 by 5.2×, with only a small recall drop.

Target-Aware Retrieval vs. All-Detected Outputs. To isolate whether target-aware retrieval drives the accuracy gain, we compare mmPointRet against a question-agnostic YOLO-box wedge filter, which retains the points falling in the wedges of *all* detected objects in the frame rather than a single referent, and against the LiDAR GT reference, which upper-bounds target-aware selection (Tab. 3(b)). mmPointRet outperforms the YOLO wedge in accuracy (39.04% vs. 37.62%) and recall (65.73% vs. 45.80%) while retaining fewer points (26.6 vs. 36.0). Moreover, despite the inherent clutter of its 2D wedge, it remains competitive with the LiDAR GT (42.72%). A question-agnostic control on the full benchmark further shows that the improvement stems from selection rather than compression.

Depth Ambiguity and LiDAR-Free Supervision. Our 2D wedge is unbounded along the line of sight and therefore admits background clutter, so we first test whether a monocular-estimated 3D box resolves this (Tab. 4). Keeping the box selected by mmPointRet, we predict its representative depth with Depth Anything V2 (Yang et al. 2024), back-project it to a 3D center, and retrieve the radar points inside a class-conditioned 3D box placed there. Because absolute range is difficult to infer from a single image, the estimated box rarely lands on the referent: it retains 0.19 points per frame against the 4.4 of the LiDAR box and recovers only 2.03% recall, degrading accuracy to 41.65%. The depth-unbounded wedge is the more robust design. We further investigate whether the costly LiDAR-derived ground-truth supervision can be replaced with pseudo-labels generated by a monocular depth model during training. DINORet replaces the LiDAR-derived supervision with 2D boxes from the open-vocabulary detector Grounding DINO (Liu et al. 2023). It reaches 46.30%, above the vision-only baseline of Qwen2.5-VL-72B (45.96%, Tab. 2) and approaching LiDAR-supervised mmPointRet (47.80%).

Effectiveness of Physical Attributes in Radar

To measure how much each attribute contributes in radar modality, we remove the attribute expected to matter for each subset from the radar text—Doppler for motion-related questions and intensity for visibility-limited scenes—and additionally remove both, holding every other component fixed (GPT-5-mini, Tab. 5). Removing Doppler consistently degrades performance on motion-related questions: accuracy drops from 40.41% to 35.23% on numeric speed questions and from 51.22% to 47.80% on qualitative ones, with a larger drop of 8.78 pp on the latter when both attributes are removed. Removing intensity likewise degrades visibility-limited scenes, from 60.31% to 55.67% in adverse weather and from 50.00% to 45.70% in micro-mobility. These results indicate that Doppler supplies temporal motion cues for speed-related reasoning while intensity supplies complementary evidence when visual cues are degraded or target agents are small and less salient, showing that the two attributes contribute to radar-vision-language reasoning in complementary ways across question types.

Conclusion

In this paper, we introduced a radar-vision-language reasoning framework for autonomous driving QA, enabling VLMs to leverage mmWave radar cues together with visual observations. Directly feeding dense radar point clouds into VLMs introduces inefficiency and noise. To address this, we proposed mmPointRet, a question-guided mmWave radar point retriever that selects target-relevant radar points for efficient and physically informed reasoning. Since the retrieved radar points are converted into a language-compatible representation, our framework can be applied to various VLMs in a model-agnostic manner. We constructed mmDrive-QA, a radar-vision-language QA benchmark designed for decision-critical driving scenarios. Experimental results show that incorporating mmWave radar information improves VLM-based driving QA. mmPointRet reduces unnecessary radar inputs while preserving question-relevant physical cues, providing consistent gains across various VLM scales and architectures. While the current pipeline relies on vision-based detectors, future work will explore radar-guided region proposals to ensure robust retrieval even under severe visual degradation. We hope this work encourages future research on language reasoning grounded in multi-sensor signals for autonomous driving beyond vision-centric models.

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